

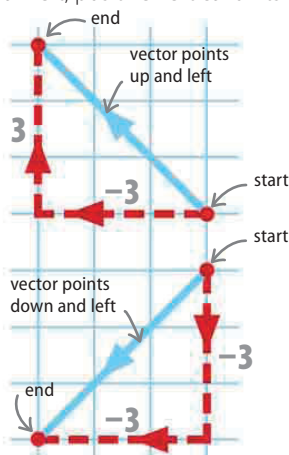
Direction of vectors

The direction of a vector is determined by whether its units are positive or negative. Positive horizontal units mean movement to the right, negative horizontal units mean left; positive vertical units mean movement up, and negative vertical units mean down.

▷ Movement up and left

This movement has a vector with negative horizontal units and positive vertical units.

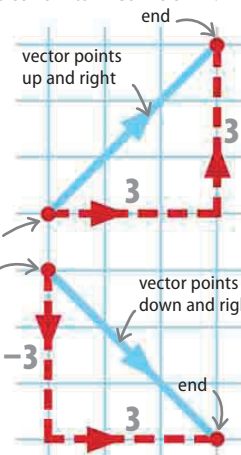
negative horizontal units mean move left $\begin{pmatrix} -3 \\ 3 \end{pmatrix}$ positive vertical units mean move up



▷ Movement up and right

This movement has a vector with both sets of units positive.

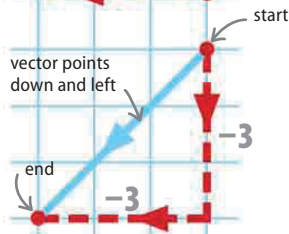
positive horizontal units mean move right $\begin{pmatrix} 3 \\ 3 \end{pmatrix}$ positive vertical units mean move up



▷ Movement down and left

This movement has a vector with both sets of units negative.

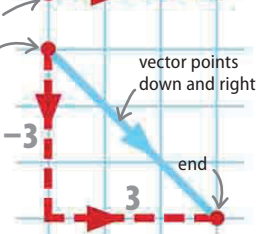
negative horizontal units mean move left $\begin{pmatrix} -3 \\ -3 \end{pmatrix}$ negative vertical units mean move down



▷ Movement down and right

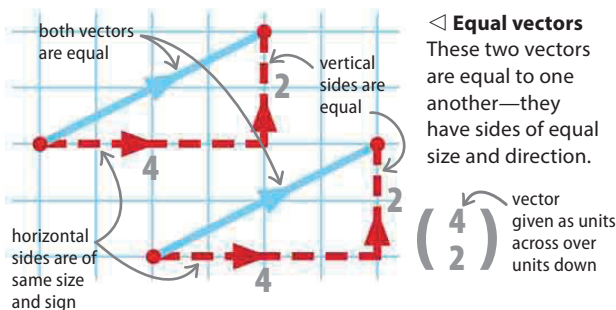
This movement has a vector with positive horizontal units and negative vertical units.

positive horizontal units mean move right $\begin{pmatrix} 3 \\ -3 \end{pmatrix}$ negative vertical units mean move down



Equal vectors

Vectors can be identified as equal even if they are in different positions on the same grid, as long as their horizontal and vertical units are equal.



◁ Equal vectors

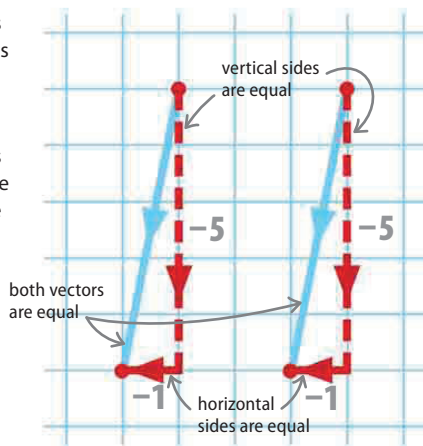
These two vectors are equal to one another—they have sides of equal size and direction.

vector given as units across over units down $\begin{pmatrix} 4 \\ 2 \end{pmatrix}$

▷ Equal vectors

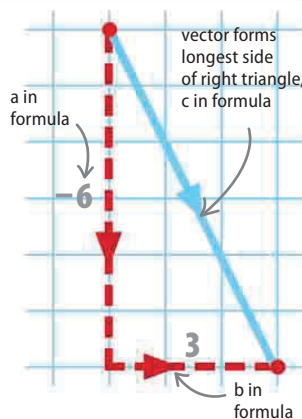
These two vectors are equal to one another because their horizontal and vertical sides are each the same size and have the same direction.

numerical expression of both vectors $\begin{pmatrix} -1 \\ -5 \end{pmatrix}$



Magnitude of vectors

With diagonal vectors, the vector is the longest side (c) of a right triangle. Use the Pythagorean theorem to find the length of a vector from its vertical (a) and horizontal (b) units.



formula for Pythagorean theorem

$$a^2 + b^2 = c^2$$

$$(-6)^2 + 3^2 = c^2$$

$$(-6)^2 = -6 \times -6 = 36$$

$$3^2 = 3 \times 3 = 9$$

$$36 + 9 = c^2$$

c^2 is the square of vector

$$45 = c^2$$

square root of 45

c is equal to the square root of 45

$$c = \sqrt{45}$$

length of vector

$$c = 6.7$$

Put the vertical and horizontal units of the vector into the formula.

Find the squares by multiplying each value by itself.

Add the two squares. This total equals c^2 (the square of the vector).

Find the square root of the total value (45) by using a calculator.

The answer is the magnitude (length) of the vector.